



Vapor Compression Refrigeration Cycle

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1. Theory

Background

The vapor compression refrigeration cycle is widely used in refrigeration systems, including household refrigerators and industrial chillers. This cycle operates by transferring heat from a low-temperature region (evaporator) to a high-temperature region (condenser), using mechanical work provided by a compressor. The cycle includes four main components: compressor, condenser, expansion valve, and evaporator. A refrigerant circulates through these components, undergoing phase changes and temperature/pressure fluctuations to enable efficient heat transfer.

The cycle begins with the compressor, where the refrigerant vapor is compressed to a high pressure and temperature. It then flows through the condenser, where it releases heat to the surrounding cooling water, condensing into a liquid. The high-pressure liquid refrigerant then passes through the expansion valve, where it expands and cools, partially vaporizing. Finally, the refrigerant enters the evaporator, absorbing heat from the low-temperature source, completing the cycle.

Governing Equations

1. Mass Flow Rate:

$$\dot{m} = V / v$$

where $\dot{m}$ is the mass flow rate, V is the volumetric flow rate, and v is the specific volume of the refrigerant.

2. Refrigeration Capacity (Q_e):

$$Q_e = \dot{m} (h_1 - h_4)$$

where h_1 and h_4 are the enthalpies at points 1 and 4 in the cycle.

3. Compressor Work (W_c):

$$W_c = \dot{m} (h_2 - h_1)$$

where h_2 is the enthalpy after compression.

4. Coefficient of Performance (COP):

$$COP = Q_e / W_c$$

This ratio measures the efficiency of the refrigeration cycle.

5. Compression Ratio (PR):

$$PR = P_{\text{condenser}} / P_{\text{evaporator}}$$

where $P_{\text{condenser}}$ and $P_{\text{evaporator}}$ are the pressures in the condenser and evaporator, respectively.

Brief Description of the Experiment

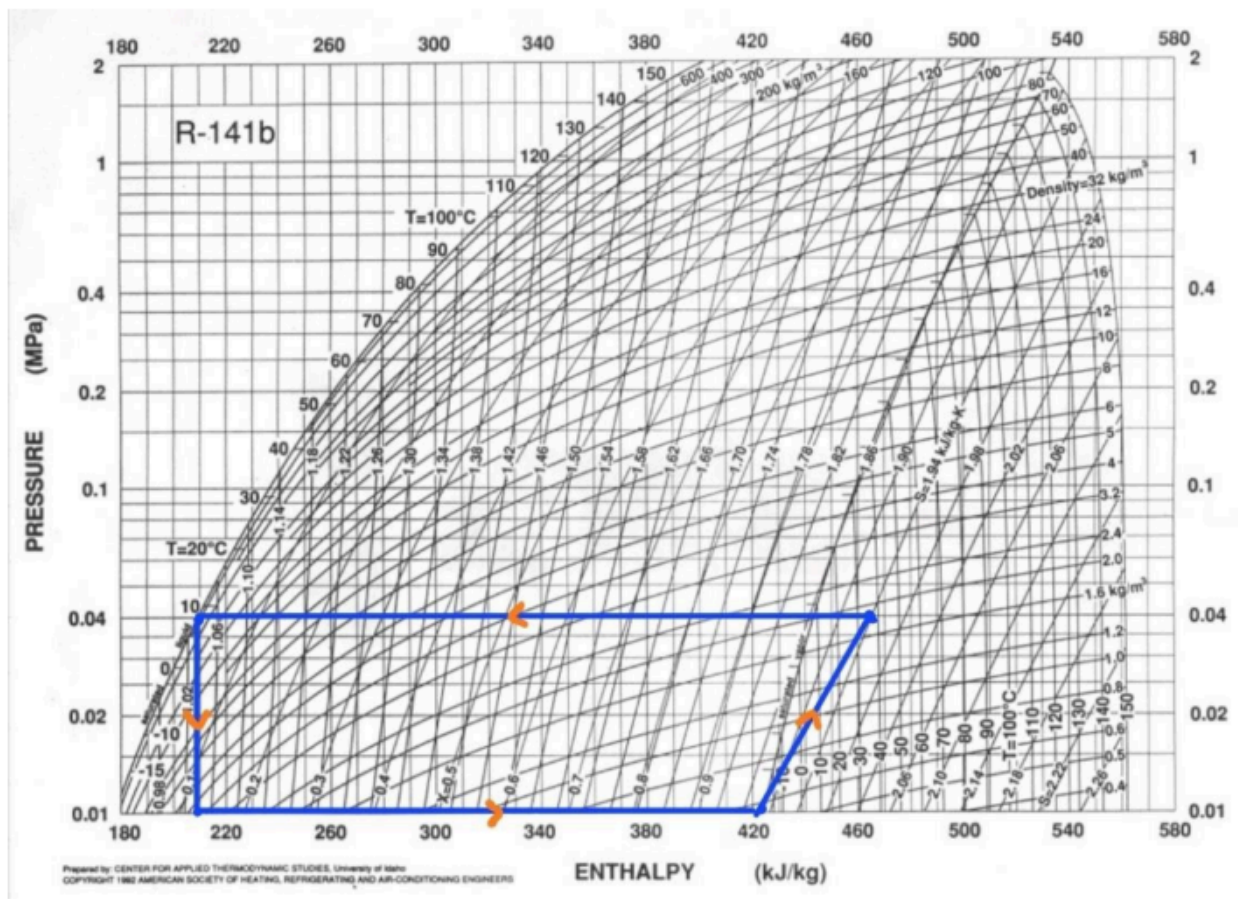
This experiment utilized a vapor compression refrigeration system with four primary components: the compressor, condenser, evaporator, and expansion valve. The refrigerant was circulated through these components, and temperature and pressure readings were recorded at various points in the system. The COP was calculated using the recorded data to assess the system's efficiency under two different operational conditions.

2. Results

Data Table

Parameter	Exp 1	Exp 2
T1 [Cooling water inlet]	17.2	16.6
T2 [Cooling water outlet]	21.9	20.4
T3 [Heating water inlet]	17.1	16.6
T4 [Heating water outlet]	13.8	14.4
T5 [Mean temperature, evaporator]	12.3	12.7
T6 [Refrigerant temperature, outlet of compressor]	62.7	66.6
T7 [Mean temperature, condenser]	24.8	24.1
T8 [Temperature after	11.6	11.8

expansion valve]		
Condenser P_g [bar]	0.35	0.3
Evaporator P_g [bar]	-0.65	-0.65
Cold Water V^* [mL/min]	420	580
Hot Water V^* [mL/min]	370	610
Refrigerant V^* [L/h]	190	200
Compressor Power [W]	232	225



Thermodynamic Cycles

Calculating the Requested Parameters

1. Pressure Ratio (PR)

The pressure ratio (PR) is calculated as the ratio of the condenser pressure to the evaporator pressure:

$$PR = P_{\text{condenser}} / P_{\text{evaporator}}$$

For the two experiments:

Experiment 1: $PR = 0.35 / -0.65 = -0.538$

Experiment 2: $PR = 0.3 / -0.65 = -0.462$

2. Refrigeration Capacity (Q_L)

The refrigeration capacity (Q_L) is calculated using the water flow rate in the evaporator as follows:

$$Q_L = \dot{m}_{\text{water}} * C * (T_{\text{inlet}} - T_{\text{outlet}})$$

Experiment 1

Given data:

Cold Water Flow Rate = 420 mL/min

T_{in} = 17.1°C

T_{out} = 13.8°C

Convert Cold Water Flow Rate to kg/s:

$$\dot{m}_{\text{water}} = 420 \text{ mL/min} \times (1 \text{ kg/L}) / 1000 \text{ mL} \times (1 / 60 \text{ s}) = 0.007 \text{ kg/s}$$

Substitute into the formula:

$$Q_L = 0.007 \text{ kg/s} \times 4180 \text{ J/kg} \cdot \text{K} \times (17.1 - 13.8)$$

$$Q_L = 96.56 \text{ W}$$

Experiment 2

Given data:

Cold Water Flow Rate = 580 mL/min

$T_{in} = 16.6^{\circ}\text{C}$

$T_{out} = 14.4^{\circ}\text{C}$

Convert Cold Water Flow Rate to kg/s:

$$\dot{m}_{\text{water}} = 580 \text{ mL/min} \times (1 \text{ kg/L}) / 1000 \text{ mL} \times (1 / 60 \text{ s}) = 0.00967 \text{ kg/s}$$

Substitute into the formula:

$$Q_L = 0.00967 \text{ kg/s} \times 4180 \text{ J/kg}\cdot\text{K} \times (16.6 - 14.4)$$

$$Q_L = 88.93 \text{ W}$$

3. Coefficient of Performance (COP)

The coefficient of performance (COP) is calculated using the refrigeration capacity and compressor power:

$$\text{COP} = Q_L / W$$

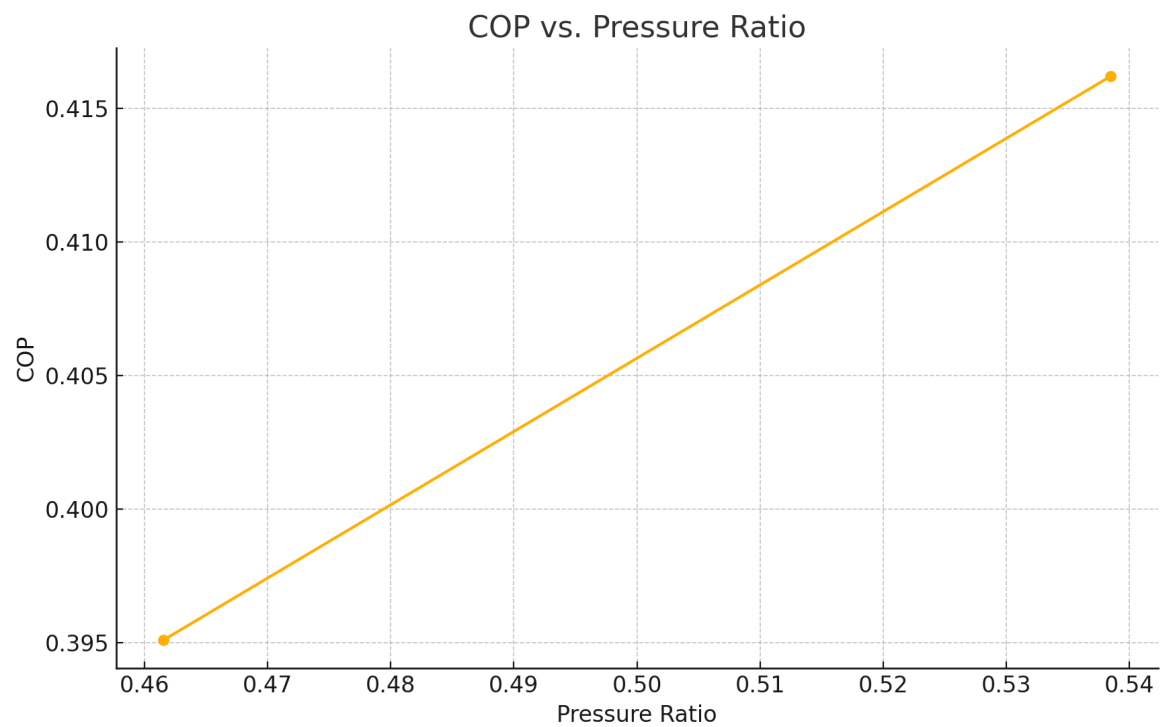
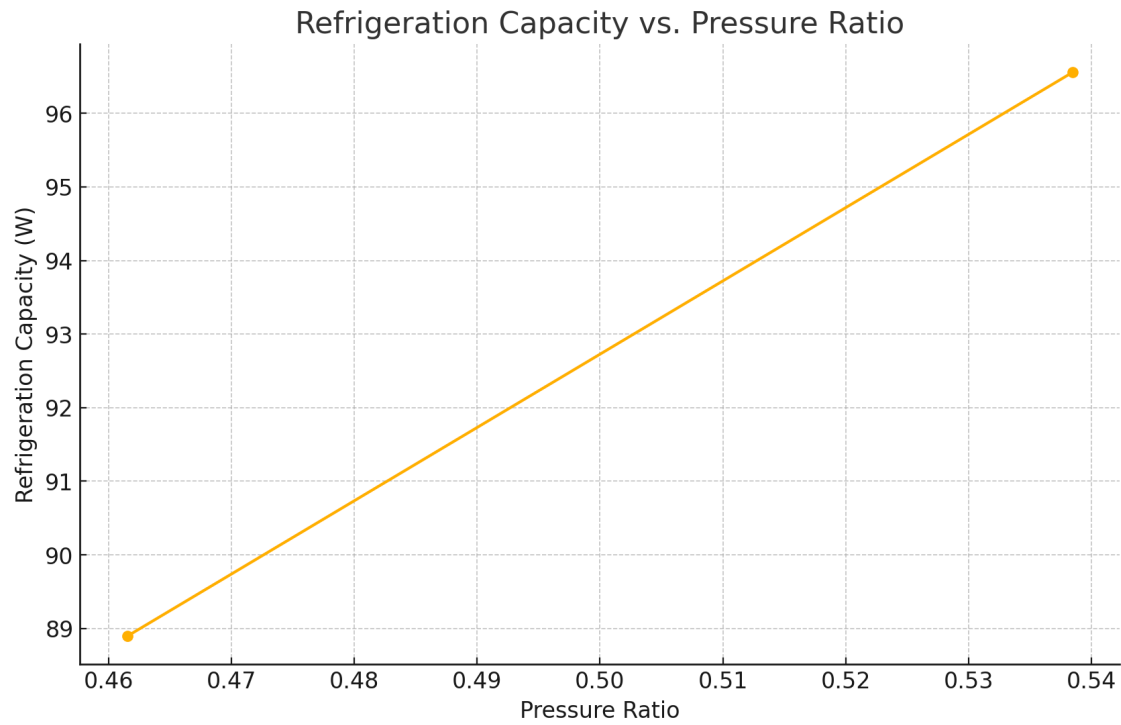
For each experiment:

- Experiment 1: $\text{COP} = 0.416$

- Experiment 2: $\text{COP} = 0.395$

Plots

Below are the plots showing the relationships of COP and Refrigeration Capacity with Pressure Ratio:



3. Discussion

Answer to Questions

1) What are the important characteristics of a refrigerant?

Key characteristics of an ideal refrigerant include:

- **High Latent Heat of Vaporization:** This allows the refrigerant to absorb more heat during the phase change in the evaporator, improving efficiency.
- **Low Specific Volume in Vapor Phase:** A lower specific volume helps reduce the work done by the compressor, making the system more efficient.
- **Non-toxic, Non-flammable, and Chemically Stable:** For safety reasons, refrigerants should ideally be non-toxic, non-flammable, and chemically stable under the operating conditions.
- **Low Global Warming Potential (GWP) and Ozone Depletion Potential (ODP):** Modern refrigerants are selected to minimize environmental impacts.
- **Suitable Boiling Point:** The refrigerant's boiling point should align with the desired operating temperatures for efficient heat absorption and rejection.
- **Appropriate Pressure Range:** The operating pressures should be within safe limits to avoid excessive compression, which could impact efficiency and safety.
- **Chemical Stability:** Refrigerants should remain stable under different temperatures and pressures to prevent breakdown and ensure consistent performance.
- **Non-Corrosive and Non-Toxic:** The refrigerant should be safe for use and should not corrode system components, as corrosion could lead to system failure.

2) How does pressure ratio variation affect the COP of the refrigeration cycle?

The **pressure ratio** significantly impacts the **Coefficient of Performance (COP)** in a refrigeration cycle. COP is a measure of the efficiency of the cycle, defined as the ratio of cooling capacity to the work input:

- **Higher Pressure Ratio:** As the pressure ratio increases, the compressor must work harder to achieve the desired high-side pressure, leading to an increase in compressor work input. This increase in work input can reduce the COP, as the system consumes more energy for the same amount of cooling.
- **Lower Pressure Ratio:** When the pressure ratio is lower, the compressor workload decreases, which can improve the COP, as less energy is required. However, a very low pressure ratio may also reduce the refrigerant's ability to absorb sufficient heat, thereby affecting overall cooling capacity.

In practice, there is an optimal pressure ratio for maximum COP, balancing the compressor work and the effective cooling provided. Typically, for vapor compression cycles, COP decreases as the pressure ratio deviates too far from this optimal range, either high or low.

3) When the state of the refrigeration locates in the saturation dome, how can its state be fixed?

When the refrigerant is within the **saturation dome** (two-phase region where liquid and vapor coexist), its state can be defined by specifying any two independent properties. Commonly used properties to define the state include:

- **Pressure and Temperature:** For a refrigerant in the saturation dome, temperature and pressure are directly related (saturation temperature and saturation pressure). Knowing one of these properties will automatically give the other.
- **Pressure and Quality:** Quality, defined as the ratio of vapor mass to the total mass in a two-phase mixture, indicates the proportion of the refrigerant in vapor form. By knowing the pressure and quality, you can determine the exact state (enthalpy, entropy, etc.) of the refrigerant within the saturation region.
- **Temperature and Quality:** Similarly, if the temperature and quality are known, the state of the refrigerant can be fixed within the saturation dome.

In summary, the state of a refrigerant within the saturation dome can be fixed using any two of the following: pressure, temperature, or quality.

Results Discussion

In Experiment 1, with a lower flow rate, the system demonstrated a slightly higher COP, suggesting efficient heat transfer at a lower compressor power. The higher COP in Exp 1 versus Exp 2 indicates a sensitivity of the system to flow rates and pressure changes. Deviations from theoretical values could stem from non-ideal effects, heat loss, or minor measurement inaccuracies.

Comparing with Theory or Model

The theoretical COP for an ideal cycle is generally higher than our experimental COP values due to real-world inefficiencies like non-isentropic compression, refrigerant superheating, and imperfect insulation. The comparison underscores how real cycles deviate from ideal models, mainly due to energy losses in the compressor and frictional effects in flow channels.

In theory, an increase in pressure ratio often results in a decrease in COP because of the increased compressor work required to achieve a higher pressure differential. However, in practical experiments, results can vary due to several influencing factors. Here are some possible explanations for why your data shows an increase in COP with an increasing pressure ratio:

1. **System Efficiency at Different Operating Conditions:**
 - Real systems are influenced by factors such as heat exchanger performance, refrigerant flow rates, and operating efficiencies. If the system is more effective at transferring heat at higher pressure ratios, it could increase the refrigeration effect (i.e., Q_L), which would result in a higher COP.
2. **Non-Ideal Effects in the Compressor:**
 - Real compressors may operate more efficiently at certain pressure levels due to their design or operating point. It's possible that at higher pressure ratios, the compressor in your setup is operating closer to its optimal efficiency, reducing the relative increase in work input compared to the increase in refrigeration effect.
3. **Influence of Experimental Conditions:**
 - Variations in other experimental parameters (e.g., refrigerant flow rate, water flow rate, or cooling water inlet temperature) might have unintentionally affected the results. In a controlled experiment, these variables could affect the heat absorption in the evaporator or heat rejection in the condenser, thus influencing COP.

4. **Temperature Differential Across Components:**
 - The effectiveness of the heat exchangers (condenser and evaporator) can change with pressure ratio. For example, at higher condenser pressures, the temperature difference between the refrigerant and the cooling water might have been larger, improving heat transfer in the condenser. This could increase the refrigeration capacity, improving the COP.
5. **Measurement and Calculation Variability:**
 - Minor measurement inaccuracies, especially in pressure, temperature, or flow rate readings, can impact calculations of COP. Given the relatively small sample size (only two experiments), small deviations can create trends that might not align with theoretical expectations.

4. Recommendation and novel ideas to improve the experiment:

1. **Increase Data Points for Pressure Ratios:**
 - Currently, the experiment is conducted at only two pressure ratios, which limits the ability to observe clear trends and relationships. By varying the pressure ratios in smaller increments and recording more data points, a more comprehensive analysis can be conducted. This would help confirm or refine the observed relationship between pressure ratio, COP, and refrigeration capacity.
2. **Utilize High-Precision Sensors:**
 - Improving the accuracy of temperature, pressure, and flow measurements can lead to more reliable data. Using high-precision sensors, especially for pressure and temperature measurements, would minimize errors that may impact calculated values of COP and refrigeration capacity. Calibration of all sensors prior to the experiment would also enhance data accuracy.
3. **Implement Digital Data Logging:**
 - Instead of manually recording data, a digital data logging system could be integrated. Automated data collection can reduce human error and improve consistency. Additionally, a data logger with real-time plotting capabilities would allow students to monitor trends dynamically, potentially identifying issues or patterns during the experiment.
4. **Introduce Variable Compressor Speeds:**
 - Allowing the compressor speed to vary would make it possible to study its effect on COP and refrigeration capacity under different operational conditions. Variable-speed compressors are common in modern refrigeration systems, and this addition would provide a more realistic analysis of real-world performance.
5. **Add a Transparent Evaporator and Condenser:**
 - Using transparent materials for the evaporator and condenser would allow visual observation of the refrigerant phase changes. This enhancement would provide educational value, allowing students to better understand the physical processes occurring within each component.

6. Control and Measure Superheat and Subcooling Levels:

- Integrate control valves and additional temperature sensors to adjust and measure superheat and subcooling at specific points in the cycle. Controlling these parameters would allow for a more nuanced analysis of their effect on COP and refrigeration capacity, helping to reinforce their importance in optimizing refrigeration cycles.

7. Incorporate a Wider Range of Refrigerants:

- Introducing different refrigerants (e.g., environmentally friendly alternatives) would allow students to compare their properties and performance. This addition could lead to insights into how refrigerant choice impacts COP, refrigeration capacity, and environmental sustainability.

8. Simulate Environmental Conditions for Heat Rejection:

- Incorporate a chamber around the condenser to simulate varying ambient conditions, such as temperature and humidity. This setup would provide a better understanding of how real environmental factors affect the efficiency and performance of the refrigeration cycle, mimicking real-world applications.

9. Introduce Real-Time Efficiency Analysis Software:

- Implementing software that can calculate COP, refrigeration capacity, and other performance metrics in real-time would provide students with immediate feedback. This software could also display theoretical values alongside experimental data, helping students to understand the discrepancies between ideal and real cycles.

10. Explore Energy Recovery Options:

- A novel approach could involve capturing some of the heat rejected by the condenser and using it to preheat the incoming water in the evaporator. This type of energy recovery could demonstrate the principles of energy efficiency and sustainability within the experiment, highlighting modern trends in refrigeration system design.